KEY SHEET

MATHEMATICS

1	**B**	2	**BD**	3	**ACD**	4	**B**	5	**A**	6	**D**
7	**C**	8	**2**	9	**125**	10	**1348**	11	**840**	12	**0**
13	**1**	14	**C**	15	**A**	16	**B**	17	**C**		

PHYSICS

18	**A**	19	**ABCD**	20	**AD**	21	**A**	22	**B**	23	**A**
24	**B**	25	**8**	26	**6**	27	**2**	28	**6**	29	**3**
30	**3**	31	**C**	32	**B**	33	**C**	34	**A**		

CHEMISTRY

35	**ABCD**	36	**D**	37	**ABD**	38	**A**	39	**D**	40	**D**
41	**D**	42	**8**	43	**729**	44	**9.5**	45	**2**	46	**5**
47	**7**	48	**A**	49	**D**	50	**A**	51	**A**		

SOLUTIONS
MATHEMATICS

1. Key: B

Sol: Any point on the line through $P(-1,1)$ at a distance of r from P will be $(-1+r\cos\theta, 1+r\sin\theta)$, where $\tan\theta$ is slope of the line.

If this line meets $xy=1$ in A & B, then $(-1+r\cos\theta)(1+r\sin\theta)=1$ will have PA & PB as its roots.

$r^2\sin\theta\cos\theta + r(\cos\theta-\sin\theta) - 2 = 0$

Also, for the point R, $x=-1+PR\cos\theta$ & $y=1+PR\sin\theta$.

Hence $\cos\theta=\dfrac{x+1}{PR}$ & $\sin\theta=\dfrac{y-1}{PR}$

Now as PA, PR & PB are in A.P., hence PA+PB=2PR

$$\Rightarrow \frac{\sin\theta-\cos\theta}{\sin\theta\cos\theta}=2PR \Rightarrow \frac{\frac{y-1}{PR}-\frac{x+1}{PR}}{\frac{x+1}{PR}\cdot\frac{y-1}{PR}}=2PR \Rightarrow 2xy=x-y$$

2. Key: ABD

Sol: $P(A\cup B)=P(A)+P(B)-P(A\cap B)$

$$\frac{5}{8}=\frac{3}{8}+\frac{4}{8}-P(A\cap B) \quad \Rightarrow P(A\cap B)=\frac{2}{8}=\frac{1}{4}$$

Now $p\left(A/B\right)=\dfrac{P(A^c\cap B)}{P(B)}=\dfrac{P(B)-P(A\cap B)}{P(B)}=1-\dfrac{2}{8}.\dfrac{8}{4}=\dfrac{1}{2}$

$$2P\left(A/B^c\right)=\frac{2P(A\cap B^c)}{P(B^c)}=\frac{2(P(A)-P(A\cap B))}{1-P(B)}=4\left(\frac{3}{8}-\frac{2}{8}\right)=\frac{1}{2}\Rightarrow \ (A) \text{ is correct}$$

$$P(A/B)=\frac{P(A\cap B)}{P(B)}=\frac{1}{4}.\frac{2}{1}=\frac{1}{2}=P(B)\Rightarrow \ (B) \text{ is correct}$$

Again $P\left(A^c/B^c\right)=\dfrac{P(A^c\cap B^c)}{P(B^c)}=\dfrac{1-P(A\cup B)}{1-P(B)}=2\left(1-\dfrac{5}{8}\right)=\dfrac{3}{4}$

$$P\left(B/A^c\right)=\frac{P(B)-P(A\cap B)}{5/8}=\frac{\frac{1}{2}-\frac{1}{4}}{\frac{5}{8}}=\frac{1}{4}.\frac{8}{5}=\frac{2}{5}$$

Hence $8P\left(A^c/B^c\right)=15P\left(B/A^c\right) \quad \Rightarrow (C)$ is not correct

Again $2P\left(A/B^c\right)=\dfrac{1}{2}$ from (1) $\Rightarrow P\left(A/B^c\right)=\dfrac{1}{4}P(A\cap B)$

Hence (D) is correct.

3. Key:ACD

Sol: We have $S=\left\{(x,y):x^2+y^2\le 2024;x,y\in R\right\}$

$(2024,2024)$ doesn't satisfy the relation S

$\therefore S$ is not reflexive.

Let $(x,y)\in S.\quad \therefore x^2+y^2\le 2024$

$\Rightarrow y^2+x^2\le 2024\Rightarrow (y,x)\in R$

$\therefore R$ is symmetric.

$\left(\sqrt{2021},1\right),\left(1,\sqrt{2022}\right)\in S$ but $\left(\sqrt{2021},\sqrt{2022}\right)\notin S$

$\therefore R$ is not transitive.

4. Key: B

Sol:Multiply R_2 by z and R_3 by y and use $R_1\to R_1-R_2+R_3$

5. Key:A

Sol: $\sum x=235,\sum y=250,\sum x^2=6750,\sum y^2=6840$

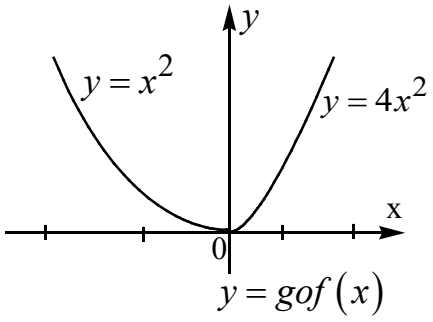
$$\Rightarrow \sigma^2=\frac{1}{10}\sum x_i^2-\left(\frac{\sum x_i}{10}\right)^2=\frac{1}{10}.6750-\left(\frac{235}{10}\right)^2$$

$$=675-552.25=122.75\Rightarrow \sigma=11.08$$

Similarly, for $\Sigma y=250,n=10,\Sigma y^2=6840$

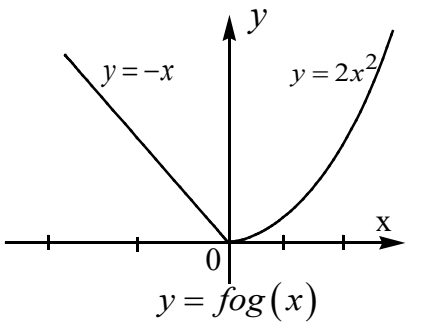
6. Key:D

Sol: $f(x)=\begin{cases}-x, & x<0\\ 2x, & x\ge 0\end{cases}$



$y=gof(x)$

Now, $gof(x)=\begin{cases}x^2, & x<0\\ 4x^2, & x\ge 0\end{cases}$



$y=fog(x)$

Similarly, $fog(x)=\begin{cases}-x, & x<0\\ 2x^2, & x\geq 0\end{cases}$ which is continuous at $x=0$ but not differentiable at $x=0$

7. Key: C

Sol: We know that in any triangle $b+c-a>0, c+a-b>0$ and $a+b-c>0$

Since, $AM\geq GM$, we get

$$\frac{(b+c-a)+(c+a-b)}{2}\geq\left[(b+c-a)(c+a-b)\right]^{1/2}$$

$$\Rightarrow c\geq\left[(b+c-a)(c+a-b)\right]^{1/2}$$

Similarly $b\geq\left[(b+c-a)(a+b-c)\right]^{1/2}$ and $a\geq\left[(c+a-b)(a+b-c)\right]^{1/2}$

Multiplying the above equations, we get

$$abc\geq(b+c-a)(c+a-b)(a+b-c)$$

8. Sol: $f(n)=\int_0^{\pi/2}\cos^n x\cos nx\,dx=\int_0^{\pi/2}\cos^n x\cos\left((n+1)x-x\right)dx$

$$f(n)=\int_0^{\pi/2}\cos^{n+1}x\cos(n+1)x\,dx+\int_0^{\pi/2}\left(\cos^n x\sin x\right)\sin(n+1)x\,dx$$

$$f(n)=f(n+1)-\left[\frac{\sin(n+1)x\cos^{n+1}x}{n+1}\right]_0^{\pi/2}+\int_0^{\pi/2}\cos^{n+1}x\cos(n+1)x\,dx$$

$$f(n)=2f(n+1)\Rightarrow\frac{f(n+1)}{f(n)}=\frac{1}{2}$$

9. Key: 125

Sol: Vertices of the tetrahedron are $(6,6,-6),(6,-6,6),(-6,6,6)\ \&\ (0,0,0)$

Two adjacent vectors representing the base are $12j-12k$ & $12i-12k$.

Area= $\frac{1}{2}\left|\left(12\hat{j}-12\hat{k}\right)\times\left(12\hat{i}-12\hat{k}\right)\right|=72\begin{Vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 0 & 1 & -1\\ 1 & 0 & -1\end{Vmatrix}=72\sqrt{3}$

10. Key: 1348

Sol: $a+c=2b\Rightarrow 3b=2025\Rightarrow b=675\Rightarrow a+c=1350$

Number of positive integer solutions will be ${}^{1349}C_1=1349$ which includes $675,675$ as the solution but shouldn't be included

11. Key: 840

Sol: $E=(x+2)^{n-1}\left[1+\frac{x+1}{x+2}+\left(\frac{x+1}{x+2}\right)^2+....+\left(\frac{x+1}{x+2}\right)^{n-1}\right]$

$$E=(x+2)^{n-1}\left[\frac{1-\left(\frac{x+1}{x+2}\right)^n}{1-\frac{x+1}{x+2}}\right]=(x+2)^n\left[\frac{(x+2)^n-(x+1)^n}{(x+2)^n}\right]=(x+2)^n-(x+1)^n$$

Now co-efficient of x^r in $\left[(2+x)^n-(1+x)^n\right]={}_r^nb={}^nC_r2^{n-r}-{}^nC_r={}^nC_r\left(2^{n-r}-1\right)$

Hence ${}_7^{10}b={}^{10}C_7\left(2^3-1\right)=840$.

12. Key: 0, Sol: $(x-a)(x-b)(x-c)(x-d)-9=(x-r)^4$

On comparing coefficient of x^3 $a+b+c+d=4r$

13. Key: 1

Sol: Using G.M. $\le$ A.M. we get

$$\left(a^ab^bc^c\right)^{\frac{1}{a+b+c}}\le\frac{a^2+b^2+c^2}{a+b+c},\left(a^bb^cc^a\right)^{\frac{1}{a+b+c}}\le\frac{ab+bc+ca}{a+b+c}\ \&\ \left(a^cb^ac^b\right)^{\frac{1}{a+b+c}}\le\frac{ac+ba+cb}{a+b+c}$$

Hence $a^ab^bc^c+a^bb^cc^a+a^cb^ac^b\le a^2+b^2+c^2+2ab+2bc+2ca=(a+b+c)^2=1$

14. Key: c

Sol: (P) $2\sin x,\sin 2x\ \&\ 2\cos x$ are in A.P., hence $\sin 2x=\sin x+\cos x$

Now Let $\sin x+\cos x=t$, then $\sin 2x=t^2-1$, then the above equation gives

$t^2-t-1=0$ or $t=\sin x+\cos x=\dfrac{1-\sqrt{5}}{2}$

(Q) Let θ be the greatest angle, then angles will be $\dfrac{\pi}{3}-\theta,\dfrac{\pi}{3},\dfrac{\pi}{3}+\theta$

$\theta=\dfrac{1}{3}\left(\dfrac{\pi}{3}+\theta\right)\Rightarrow\theta=\dfrac{\pi}{6}$ Hence angles are $\dfrac{\pi}{6},\dfrac{\pi}{3},\dfrac{\pi}{2}$

Now $\dfrac{a}{\sin\frac{\pi}{6}}=\dfrac{b}{\sin\frac{\pi}{3}}=\dfrac{c}{\sin\frac{\pi}{2}}\Rightarrow\dfrac{a}{1}=\dfrac{b}{\sqrt{3}}=\dfrac{c}{2}$.

(R) Let $AB=a,BC=\sqrt{3}a,CD=3a\ \&\ AD=3\sqrt{3}a$

Now $\cos A+\cos C=0\Rightarrow\dfrac{a^2+27a^2-BD^2}{3\sqrt{3}a^2}+\dfrac{3a^2+9a^2-BD^2}{3\sqrt{3}a^2}=0$

$\Rightarrow BD^2=20a^2$ *or* $\dfrac{BD}{AB}=2\sqrt{5}$

$(S)\,a+c=2b\ \&\ b^2=\dfrac{2a^2c^2}{a^2+c^2}\Rightarrow(a+c)^2=\dfrac{8a^2c^2}{a^2+c^2}$

$\Rightarrow(a+c)^4-2ac(a+c)^2-8a^2c^2=0\ \Rightarrow\dfrac{(a+c)^2}{ac}=-8$

But 4 gives a=c, hence by taking -2, $\Rightarrow a^2+4ac+c^2=0\Rightarrow\dfrac{a}{c}=\sqrt{3}-2$

15. Key: a

Sol: (P) Equation of any curve through the points in which given lines meet coordinate axes will be $(a_1x-y+b)(a_2x-y+b)+\lambda xy=0$. If these points are concyclic then $a_1a_2=1$

(Q) Any point on $x^2+y^2=a_1^2$ will be $(a_1\cos\theta, a_1\sin\theta)$. The chord of contact of the tangents drawn to $x^2+y^2=4$ from this point will be $a_1x\cos\theta+a_1y\sin\theta=4$.

If this touches the circle $x^2+y^2=a_2^2$, then $a_1a_2=4$.

(R) If the two circles cut orthogonally, then $2(-a_1\times a_1-a_2\times a_2)=(-1-1)$

$\Rightarrow a_1^2+a_2^2=1$

(S) Centre of the given circle is $(2,-3)$ & radius is 5. Now distance between $(8,5)$ & the centre is 10. Hence $a_1=5$ & $a_2=15$

16. Key: b

Sol:(P) $P(A|C)+P(B|C)-P((A\cap B)|C)=\dfrac{P(A\cap C)+P(B\cap C)-P(A\cap B\cap C)}{P(C)}$

$=\dfrac{P((A\cup B)\cap C)}{P(C)}=P(A\cup B|C)$

(Q) $P((A\cap\overline{B})|C)+P((A\cap B)|C)=\dfrac{P(A\cap\overline{B}\cap C)+P(A\cap B\cap C)}{P(C)}$

$=\dfrac{P(A\cap C)}{P(C)}=P(A|C)$

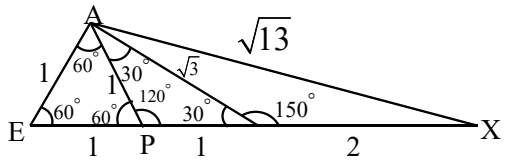
(R) $\dfrac{1-P(A)-P(B\cup C)+P(A\cap(B\cup C))}{1-P(B\cup C)}=\dfrac{1-P(A\cup(B\cup C))}{1-P(B\cup C)}=\dfrac{P(\overline{A\cup(B\cup C)})}{P(\overline{B\cup C})}$

$=\dfrac{P(\overline{A}\cap(\overline{B}\cap\overline{C}))}{P(\overline{B}\cap\overline{C})}=P(\overline{A}|(\overline{B}\cap\overline{C}))$

(S) $P(\overline{A}\cup\overline{B})+P(A\cap B)=P(\overline{A\cap B})+P(A\cap B)=1$

17. Key: c

Sol: We have $AE=EP=AP=1\Rightarrow \quad AP=PT=1$



$\Rightarrow \Delta APT$ is isosceles

This makes ΔAET a $30^\circ-60^\circ-90^\circ$ triangle,

So $AT=\sqrt{3}x$ and $\angle ATX=150^\circ$

Since TX=2 $\quad\therefore\quad$ By applying Cosine rule in ΔATX, we get

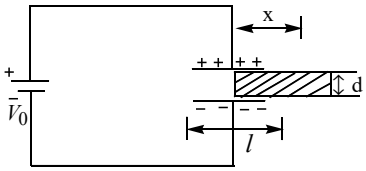
$(AX)^2=3+4-4\sqrt{3}\cos150^\circ=7+6=13\Rightarrow AX=\sqrt{13}$

$\therefore \quad \cos\angle XAE=\dfrac{13+1-16}{(2)(\sqrt{13})(1)}=\dfrac{-2}{2\sqrt{13}}=\dfrac{-1}{\sqrt{13}}$

19.

20. Under normal conditions total energy, potential energy and kinetic energy in ground state and first excited state are $-13.6\,eV$, $-27.2\,eV$, $-13.6\,eV$, $-3.4\,eV$, $-6.8\,eV$ and $3.4\,eV$ respectively. If potential energy in ground state is taken to be zero, then kinetic energy will remain unchanged but potential and total energies are increased by 27.2 eV. Therefore, the new values are 13.6 eV, 0, 13.6 eV, 23.8 eV, 20.4 eV and 3.4 eV respectively.

21. Capacity with partially inserted slab (upto depth x)



=Capacity f air capacitor + Capacity of dielectric field capacitor $= \frac{\varepsilon_0 b(l-x)}{d} + \frac{K\varepsilon_0 bx}{d}$

b=plate width $\Rightarrow C = \frac{\varepsilon_0 b}{d}\left[1 + x(K-1)\right]$ $\Rightarrow dU = \frac{1}{2}V^2(dC)$

And $(dC) = \frac{\varepsilon_0 b}{d}(K-1)dx$ $\therefore dU = \frac{\varepsilon_0 b}{2d}(K-1)V^2 dx$

Work done by force=Fdx, Increase in charge=dq $\Rightarrow dq = Vdc = \frac{\varepsilon_0 bV}{d}(K-1)dx$

This charge is transformed from negative terminal to the positive terminal inside battery.

Hence, work done by battery, $dW = -dq.V = \frac{-\varepsilon_0 bV^2(K-1)dx}{d}$

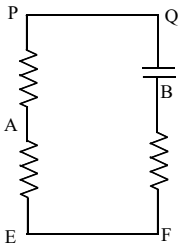
As, $dW = -dU$, Hence, $\Rightarrow F = \frac{\varepsilon_0 bV^2(K-1)}{2d}$

Clearly, force does not depend on x.

Also, acceleration of slab $a = \frac{F}{m} = \frac{\varepsilon_0 bV^2(K-1)}{2dm}$ and acceleration is also independent of dimension x.

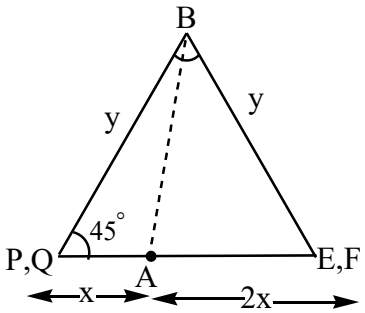
$\therefore$ Time period of oscillation is also independent of x.

22. Let voltage phasors along be PA, AE be x, 2x



And voltage phasors along QB, BF be y, y. (as $R = \frac{1}{\omega C}$).

The phasors diagram representing them can be drawn as shown



So phasor length AB (i.e; reading of V_2) is=$\sqrt{x^2+y^2-2xy\cos 45^\circ}$ (using cosine rule in ΔPAB)

Also $x+2x=V$ and $\sqrt{2y}=V \Rightarrow$ Reading of $V_2=\dfrac{V}{3}\sqrt{\dfrac{5}{2}}$

23. Let assume the average value of steady state voltage across capacitor is V. Net charge does not change over a cycle, so

$i_1\Delta t+i_2\Delta t=0 \qquad \dfrac{(\varepsilon_1-V)}{R_1}\Delta t+\dfrac{(\varepsilon_1-V)}{R_1}\Delta t=0$

$\Rightarrow V=\dfrac{\varepsilon_1 R_2+\varepsilon_2 R_1}{R_1+R_2}$ Ce=CV

24. $b\rightarrow$ radius of spot $b=a+\dfrac{\lambda L}{a}$

Geometrical spread+ spread due to diffraction

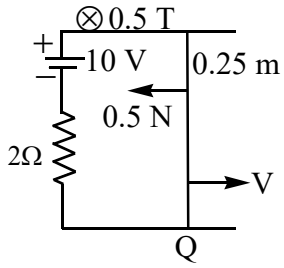
$b=a+\dfrac{\lambda L}{a} \quad \Rightarrow 1-\dfrac{\lambda}{a^2}L=0$

$\Rightarrow a^2=L\lambda \quad \Rightarrow a=\sqrt{L\lambda}$

$b_{\min}=\sqrt{4L\lambda}$

25. Induce e.m.f= $B\ell v=0.125V$

Current $I=\dfrac{10-e}{R}=\dfrac{10-0.125V}{2}$



Force $BI\ell=0.5\left(\dfrac{10-0.125}{2}\right)0.25=0.5N\left(given\right)$

Solving $v=16m/s.$

26. According to conservation of energy $\dfrac{hc}{\lambda_1}+\dfrac{hc}{\lambda_2}=RchZ^2\left(\dfrac{1}{l^2}-\dfrac{1}{n^2}\right)n=6.03$ quantum number=6

27. Let each plate moves a distance ‘x’ from its initial position

Let q charge flows in the loop. Using KVL

$$\frac{\left(\frac{Q}{2}-q\right)(d+x)}{\varepsilon_0 A}-\frac{\left(\frac{Q}{2}+q\right)(d-x)}{\varepsilon_0 A}=0$$

$$\therefore q=\frac{Qx}{2d_0};I=\frac{dq}{dt}=\frac{Q}{2d_0}\left(\frac{dx}{dt}\right)\quad I=\frac{Qv_0}{2d_0}$$

28. Refer theory

29. $E_i=\frac{k(2Q)}{(2R)^2}-\frac{k(Q)}{(4R)^2}=\frac{7kQ}{16R^2}$

30. Let K_1 & K_2 and P_1 & P_2 are the K. E and momentum of the α - particle and remaining nucleus, then $K_1+K_2=5.5MeV----(i)$

From conservation of linear momentum $P_1=P_2$

$\Rightarrow\sqrt{2K_1\times 4m}=\sqrt{2K_2\times 216m}\Rightarrow K_1=54K_2----(ii)$

From (i) and (ii)

$K_1=\frac{5.5\times 54}{55}=5.4MeV$

31.

32.

33.

34. A) $\Delta x=\frac{dy_1}{D_1}+\frac{dy_2}{D_2}$

B) $\Delta x=\frac{dy_1}{D_1}$

C) $\Delta x=\frac{dy_1}{D_1}+\frac{dy_2}{D_2}-(\mu-1)t$

D) $\Delta x=\frac{dy_1}{D_1}-(\mu-1)t$

CHEMISTRY

35. In the oxidation of oxalic acid by $KMnO_4$, the Mn^{2+} ions formed act as catalyst.

36. $Glucose \xrightarrow{dilute\ HNO_3} glucharic\ acid$

37. $\left[Ni(NH_3)_6\right]^{2+} + 3en \rightarrow \left[Ni(en)_3\right]^{2+} + 6NH_3 \left(\Delta S > 0\right)$

38. $\Delta S = \left(\dfrac{\Delta H - \Delta G}{T}\right)$

$\Rightarrow \Delta H = -nF\left[E - T\left(\dfrac{\partial E}{\partial T}\right)_P\right] and\ \Delta G = -nFE$

$\Delta S = nF\left(\dfrac{\partial E}{\partial T}\right)_P = 3 \times 69500 \times \left(\dfrac{-0.0006}{10}\right)$

$= -17.37 J / k$

39. $T_b = 80.26, \quad \Delta T_b = 0.16$

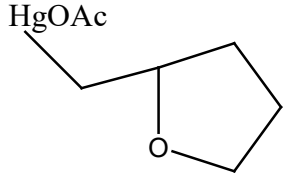
$0.16 = 2.53 \times \dfrac{0.26 / M}{11.20} \times 1000; M \approx 367$

That is almost molar mass of $C_{20}H_{16}Fe_2$.

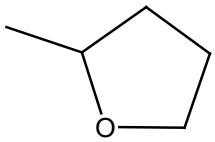
40. $P = CH_2 = CH - CH_2MgCl$

$Q = CH_2 = CH - CH_2 - CH_2 - CH_2 - OH$

R=



S=



41. The reduction of a metal oxide easier if the metal formed is in liquid state at the temperature of reduction. The entropy is higher if the metal is in liquid state than when it is in solid state. The value of entropy change (ΔS) of the reduction process is more on +Ve side when the metal formed is in liquid state and the metal oxide being reduced is in solid state. Thus the value of $\Delta_r G^\circ$ becomes more on negative side and the reduction becomes easier.

42. $cell = Ag / Ag^+ // AgI / AgI / Ag$

$Left: Ag \rightarrow Ag^+ + e^-; E^\circ = +0.799V$

$Right: AgI + e^- \rightarrow Ag^+ + I^-; E^\circ = -0.151V$

$AgI \rightarrow Ag^+ + I^-; E^\circ = -0.95V$

$E = E^\circ \dfrac{-0.0591}{n} \log Q$

$$0 = -0.9501 - \frac{0.0591}{1}\log Ksp$$

$$\log Ksp = -16.1$$

$$Ksp = 8\times10^{-17}$$

43. $$M_{AgX} = \frac{\rho \times N_A \times a^3}{4} = \frac{\frac{40}{6.022}\times 6.022\times10^{23}\times\left(6\times10^{-8}\right)^3}{4}$$

$$= 216\,g\,/\,mole$$

$$\therefore M_X = 216 - 108 = 108\,g\,/\,mole$$

If all atoms along one body diagonal have been removed, the formula of unit cell

$$= Ag_{4-1}X_{4-\frac{1}{4}} = Ag_3X_{\frac{15}{4}}$$

$$\therefore \text{ Mass of 1 mole of these unit cell } = 108\times3 + 108\times\frac{15}{4} = 729.00$$

44. 1 mole of acetone reduction to isopropanol requires 0.25 moles of $LiAlH_4$

45. (I, IV, V, X)

46. $Co^{+2}:d^7:$Sq planar complex. $\left(d^2_{yz} = d^2_{zx}\right) < d^2_{z^2} < d^1_{xy} < d_{x^2-y^2}$ (splitting of d – orbitals)

47.

$$x = \left[Fe_3(OH)_2(CH_3COO)_6\right]^+$$

$$y = \left[Fe(OH)_2(CH_3COO)\right]$$

48. For reaction $A \rightarrow product$

Using the given formula we can solve all the options

For zero order reaction $t = \frac{[A]_0 - [A]_t}{k}$

For first order reaction $t = \frac{1}{k}\ell n\frac{[A]_0}{[A]_t}$

For second order reaction $t = \frac{1}{k}\left[\frac{1}{[A]_t} - \frac{1}{[A]_0}\right]$

49. Conceptual

50.

51.

